

Supplementary Diagnostics for Representative Market-State Stock Selection

Scope

This supplement contains detailed diagnostic tables moved out of the main manuscript to keep the LNCS conference version within the 14-page limit. The k-ablation and forward-split diagnostics use the recovered main fixed-universe return matrix included in the replication package; they are intended as sanity checks rather than byte-for-byte reproductions of the saved Step 4 ranking artifact.

S1. Step 4 Momentum and Benchmark Comparison

Method	h	l	Cum.	Ann.	Sharpe	MDD	Turnover
Mapper	21	20%	401.27%	17.51%	0.970	-35.84%	16.79%
Global DBSCAN	21	20%	379.06%	16.98%	0.944	-36.62%	15.14%
PCA-KMeans	21	20%	421.88%	17.99%	0.998	-35.51%	17.23%
Equal Weight Universe	63	–	348.45%	16.21%	0.916	-37.39%	5.76%
S&P 500	21	–	242.64%	13.12%	0.773	-33.92%	0.00%
Momentum 21D	42	40%	279.00%	14.27%	0.852	-36.06%	60.30%
Momentum 63D	21	20%	305.02%	15.03%	0.858	-33.14%	47.11%
Momentum 126D	42	20%	390.98%	17.27%	0.924	-36.48%	46.99%

S2. PCA-KMeans k-Ablation

k	Cum.	Ann.	Sharpe	MDD	Turnover	Avg. stocks
1	350.03%	16.29%	0.892	-37.80%	16.99%	70.9
2	329.21%	15.74%	0.899	-36.58%	22.18%	127.8
3	409.10%	17.73%	0.991	-35.54%	17.08%	175.2
4	348.75%	16.25%	0.926	-36.80%	21.05%	210.1
5	363.85%	16.64%	0.948	-36.08%	15.32%	236.5

S3. Simple Forward-Split Diagnostic

Period	Role	h	l	Cum.	Ann.	Sharpe	MDD	Turnover
2016–2018	calibration-selected config	21	20%	50.41%	14.64%	1.114	-18.88%	17.40%
2019–2025	fixed holdout evaluation	21	20%	226.51%	18.47%	0.950	-36.16%	18.37%

S4. Daily Sharpe Bootstrap Intervals

Method	Configuration	Mean	90% CI	95% CI
<i>2020-start robustness</i>				
PCA-KMeans	$h = 63, l = 40\%, k = 3$	0.659	[0.010, 1.306]	[-0.078, 1.457]
Mapper	$h = 63, l = 40\%, k = 3$	0.644	[-0.045, 1.311]	[-0.131, 1.421]
Equal Weight	$h = 63$	0.695	[0.019, 1.383]	[-0.118, 1.517]
S&P 500	$h = 21$	0.737	[0.068, 1.382]	[-0.030, 1.577]
Momentum 21D	$h = 42, l = 40\%$	0.727	[0.067, 1.401]	[-0.119, 1.575]
<i>2023-onward diagnostic</i>				
PCA-KMeans	$h = 21, l = 40\%, k = 3$	0.910	[0.013, 1.801]	[-0.120, 1.981]
Mapper	$h = 42, l = 40\%, k = 3$	0.793	[-0.104, 1.741]	[-0.227, 1.903]
Equal Weight	$h = 63$	0.895	[-0.000, 1.824]	[-0.133, 1.996]
S&P 500	$h = 21$	1.416	[0.553, 2.370]	[0.417, 2.577]
Momentum 126D	$h = 63, l = 40\%$	0.928	[0.038, 1.837]	[-0.086, 2.085]